AYUSH GUPTA

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EDUCATION

Course	College / University	Year	CGPA / %
B. Tech (Electrical Engineering)	Netaji Subhas University of Technology, Delhi	2027	7.73
CBSE (Class XII)	Navyug Senior Secondary School, Delhi	2022	86%
CBSE (Class X)	DAV Public School, Delhi	2020	95%

EXPERIENCE

Software Engineer Intern | Bachatt (Fintech) *May 2026 – July 2026*

- Owned the **Atlassian → self-hosted migration** end-to-end — deployed Plane + Outline on **Azure AKS** (Helm charts, ingress-nginx, cert-manager, TLS) and wrote an **idempotent Python Jira→Plane REST importer** handling statuses, assignees, comments and sub-tasks, so a re-run never duplicates. Live at plane.bachatt.app / wiki.bachatt.app — then shipped features into the forks themselves, including single-assignee enforcement in the tracker backend and inline PDF rendering with a selectable text layer.
- Built **plane-slack-relay** from scratch (Node.js) — a two-way Plane↔Slack bridge over **Socket Mode** with DM-threaded ticket notifications, a per-ticket in-progress timer and scheduled sprint digests, replacing manual status-chasing for a 10+ person team.
- Stood up production infrastructure on **Azure AKS**: a shared Helm chart (Deployment + Service + NGINX Ingress + **HPA**), cluster bootstrap script, in-cluster Redis and Kubernetes Secrets management, deployed through Jenkins.
- Closed a **2x app load-time regression** after a CDN migration — compressed **134 live assets** for **~47 MB freed at ~75% average reduction**, republished to AWS S3 and Azure Blob — then made it permanent with an enforced **100 KB image gate** and **zip central-directory scanning** to close the archive bypass without extracting untrusted entries.
- Wrote two **Python data pipelines**: a daily credit-card MIS funnel report normalising ZET, Uni and Bajaj into one scheduled workbook, and a scraper aggregating four public sources (AMFI, SEBI, NISM, NSDL) plus an email intake into **Postgres**, deduped on a natural key — with **OCR on event banners** to recover speaker, topic and date from image-only invites.

Brand Marketing Intern | CashKaro *Mar 2026 – Apr 2026*

- Instrumented an **install → signup → first-transaction funnel** for a campus growth pilot, modelled per-user unit economics and converted **40+ users in under 3 weeks**.

PROJECTS

Project Aether — Offline-First Edge-AI Mesh Network | *Python, FastAPI, Node.js, libp2p, React* GitHub

- Built a disaster-response network that assumes the internet is gone: a **quantized SLM runs on-device** to extract hazard reports from local text and images offline, parsed strictly against a **pydantic** schema shared across the Python, Node and React layers.
- Chose a **grow-only-set CRDT** so merge is commutative, associative and idempotent — two nodes reconcile correctly whenever they meet, in any order, after any gap. Identity is **(node_id, seq)** and ordering is a **Lamport clock**: offline devices have no NTP, so wall-clock time is never merge order.
- Implemented the P2P transport on **libp2p** (mDNS discovery, gossipsub, anti-entropy catch-up): three nodes converge from cold start in **~2s**, and a real network partition provably diverges then **reconverges exactly in ~1s** on heal.
- Ran a **chaos matrix against the real code** — clock skew, duplicate injection, concurrent ingest, SIGKILL mid-write, hung peer — which found and fixed **two genuine defects** the 35-test unit suite missed: a thread-safety race on ingest and a missing read timeout on the sync path.

Findesk — Equity Research & Strategy Platform | *Next.js, TypeScript, Python, Supabase* Live

- Shipped **0-to-1 to six live surfaces** (research terminal, mechanical screen, portfolio tracker, paper-trading desk, compare, watchlist) on Supabase auth with **row-level security** and **89 unit tests**.
- Built a portfolio-aware assistant on **Gemini function calling** with 4 server-side tools; user identity comes from the **session cookie and never the request body**, so a crafted request cannot read another user's holdings, with a deterministic fallback when the model quota is exhausted.
- Backtested a value + momentum strategy over a **715-stock NIFTY-500 universe**, made **survivorship-free** by reconstructing point-in-time index membership from **9 archived snapshots (2017–26)**, so delisted names stay in each cohort's pool.

Scrum Dashboard — Sprint Health on Self-Hosted Plane | *Next.js, TypeScript, Prisma, Postgres* Live

- Split identity from authorisation: Google SSO answers who you are, while each user links their own Plane token — validated before storage and **encrypted at rest** — so the dashboard shows exactly what that user's permissions allow. Read-only against Plane; webhooks verified by **HMAC-SHA256**.
- Designed the snapshot cron around an irreversible constraint: a missed day is a permanent hole in the burndown, so charts render gaps as gaps rather than interpolating, and a repeat run is a no-op via a **(cycleId, date)** constraint.

POSITIONS OF RESPONSIBILITY

- **Executive Committee, PM Club (NSUT)**, Sep 2025 – Present — run weekly product case sessions for **50+ members**, teaching prioritisation and metrics frameworks to juniors.
- **Logistics Coordinator, Moksha & Resonanz (NSUT)**, Mar 2024 – Present — ran live operations for **50+ events across 6 venues** at North India's largest college festival — 20+ volunteers, 15+ vendors, **zero escalations**.

SKILLS

Languages: Python, TypeScript/JavaScript, C++, SQL

Frameworks & Backend: FastAPI, Node.js, Next.js, React, REST APIs, Socket.io, Prisma, PostgreSQL, MongoDB, Supabase, BigQuery

Infra & DevOps: Docker, Kubernetes (Azure AKS), Helm, NGINX Ingress, cert-manager, HPA, Redis, Jenkins CI/CD, GitHub Actions, Azure Blob, AWS S3, Vercel, Git

Distributed Systems: CRDTs (grow-only sets, merge semantics), libp2p (mDNS, gossipsub, anti-entropy), Lamport clocks, partition tolerance, idempotent job design

Data, AI & Testing: pandas, NumPy, statsmodels, scikit-learn, Streamlit, web scraping & OCR, LLM function/tool calling, local quantized inference (Ollama), pydantic validation, pytest, property-based and chaos testing, CI suites

ACHIEVEMENTS & CERTIFICATIONS

- Solved **600+** data-structures and algorithms problems on **LeetCode**, **Pupil on Codeforces**.
- Co-authored a **peer-reviewed research paper** on solar-cell performance enhancement — *Physica Scripta* (IOP Publishing). **Gold Medallist**, NPTEL Programming in Java (IIT-certified).